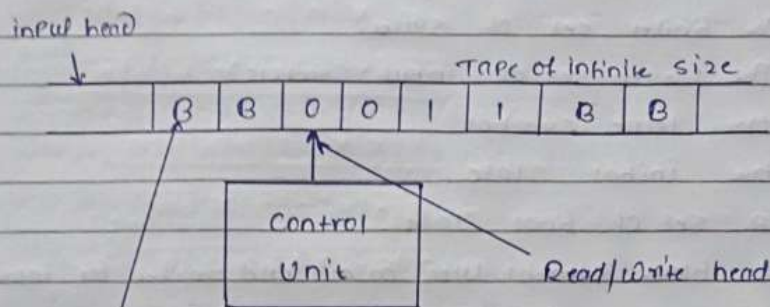


Section-A (2*5=10)

1. Draw the block diagram of Turing machine.

Ans Block diagram of Turing machine -



Default symbol

of Tape: Blank

- Tape: An infinite tape divided into cells (where each cell hold symbol from the alphabet).
- Head: A Read/Write head
- Control unit:- Determines the operation.
- Finite set of states (Q):- Represent the states of the machine.

Q2. Differentiate T.M and Push-Down Automata.

Ans Features	Push-Down Automata	Turing Machine
• Tape	Has only one tape.	Has infinite tape
• Memory	Has a stack	Has infinite memory
• Computational power	Less powerful	More powerful
• Acceptance of languages	Accepts context-free languages	Accepts all languages.
• Type of Automata	Non-deterministic	Deterministic

Q3. What are the components of T.M.?

Ans. A Turing machine can be defined as a collection of 7 components:

- Q: The finite set of states.
- Σ : The finite set of input symbols.
- Γ : the tape symbol.
- q_0 : the initial state.
- F: A set of final state.
- B: A blank symbol used as an end marker for input.
- δ : Transition function.

Q4. Define Move of a Turing Machine.

Ans. A move of a T.M. refers to the actions the machine performs during a single step of its computation. These actions are dictated by the machine's current state and the symbol it reads from the tape. A move consists of three components:

- (i) write
- (ii) Move the Tape Head
- (iii) State Transition

• Formal Representation:-

If the Turing machine's current state is q , and the symbol being read is a , a move is determined by the transition function:-

$$\delta(q, a) = (q', b, d)$$

where:-

- q' : The new state after the move.
- b : The symbol to write on the tape.
- d : The direction to move the tape head (Left L, Right R or None).

5. What is T.M. Computable?

Ans. A function is Turing machine Computable if there exists a Turing machine that produces the desired o/p for every valid i/p.

• Section-B (7*3=21)

1. Draw the computation sequence of the input string 00, using below TM.

State	Tape Symbol		
	b	0	1
→q ₁	IL q ₂	OR q ₁	
q ₂	bR q ₃	OL q ₂	IL q ₂
q ₃		bR q ₄	bR q ₅
q ₄	OR q ₅	OR q ₄	IR q ₄
(q ₅)	OL q ₂		

Ans. For the input string 00b, we get the following sequence.

q₁ 00b ⊢ 0q₁ 0b ⊢ 00q₁ b ⊢ 0q₂ 01
 ⊢ q₂ 001 ⊢ q₂ b 001 ⊢ b q₃ 001
 ⊢ b b q₄ 01 ⊢ b b 0 q₄ 1 ⊢ b b 0 1 q₄ b
 ⊢ b b 0 1 0 q₅ ⊢ b b 0 1 q₂ 00 ⊢ b b 0 q₂ 1 00
 ⊢ b b b 1 q₄ 00 ⊢ b b b 1 0 q₄ 0 ⊢ b b b 1 0 0 q₄ b
 ⊢ b b b 1 0 0 0 q₅ b ⊢ b b b 1 0 0 q₂ 00 ⊢ b b b 1 0 q₂ 0 00
 ⊢ b b b 1 q₂ 0 0 0 0 ⊢ b b b q₂ 1 0 0 0 ⊢ b b q₂ b 1 0 0 0
 ⊢ b b b q₃ 1 0 0 0 ⊢ b b b b q₅ 0 0 0 0

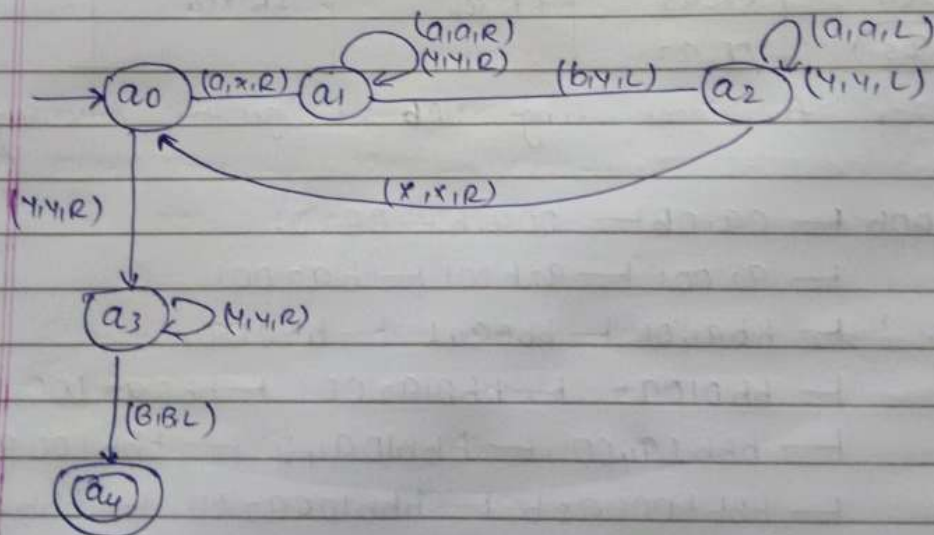
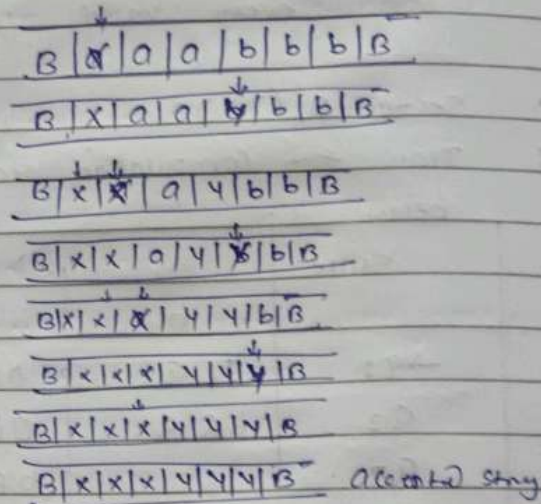
Since the Machine halts in q₅ (final state)

Q2. Construct a TM that accept $L = \{a^n b^n \mid n \geq 1\}$

Ans $L = a^n b^n \quad n \geq 1$

So $ab, aabb, aaabbb, \dots$

So string = $aaabbb$



Q3. What is Turing Machine? Explain its tuple.

Ans A Turing machine is a theoretical model of computation that manipulates symbols on infinite tape according to a set of rules.

OR

A Turing machine is a mathematical model which consists of an infinite length tape divided into cells

on which input is given. It consists of a head which reads the input tape. A state register stores the state of the T.M. After reading an input symbol, it is replaced with another symbol, its internal state is changed, and it moves from one cell to the right or left. If the TM reaches the final state the input string is accepted, otherwise rejected.

A Turing machine expressed as a 7-tuple which is: -

- Q : A finite set of states.

T : The tape alphabet

B : Blank symbol

Σ : The input alphabet

δ : a Transition function which maps $Q \times T \rightarrow Q \times \{L, R\}$.

q_0 : The initial state

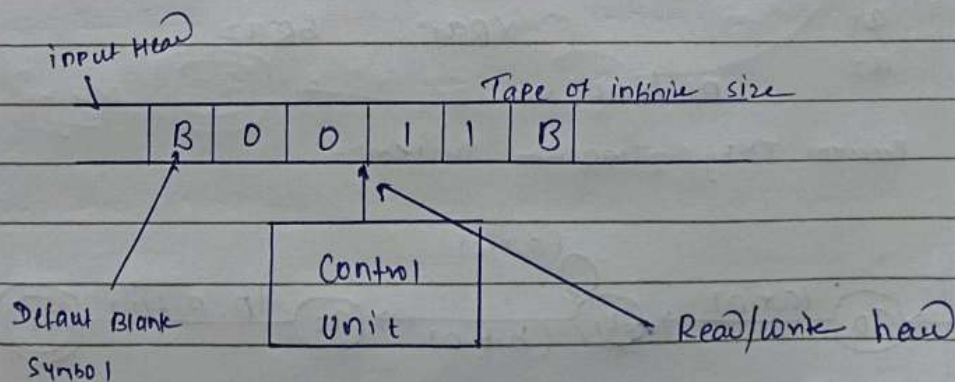
F : A set of final states.

- Section-C (3x11=33)

1. Explain block diagram of TM. Discuss following question as well with the help of example.

- Represent Turing machine through ID (instantaneous description)
- Represent T.M by transition table.
- Represent T.M by transition system.

Ans



- Tape: An infinite tape divided into cells where each cell hold symbol from the alphabet.
- Head: A read/write head that moves left or right on the tape, reading symbol & new one.
- Control Unit: contains a finite set of state and a transition function.

(a) Represent TM through ID.

ID of a TM represent its current configuration. it includes (i) current state (ii) Tape content (iii) position of Head.

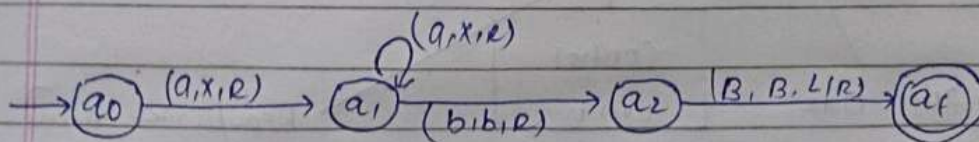
ex: I/P string : aabb

- initial ID ($q_0, -aabb-$)
- After Transition ($q_1, -xabb-$)
- final ID ($q_f, -xxbb-$)

(b) Represent TM by transition table:

State	Input Symbol		
	B	a	b
q_0		xRa_1	
a_1		xRa_1	bRa_2
a_2	BL/raf		

(c) Represent TM by Transition System:-



2. Write the tuple of T.M. Construct TM for the following:
 $L = \{ wwr \mid w = (a+b)^* \}$

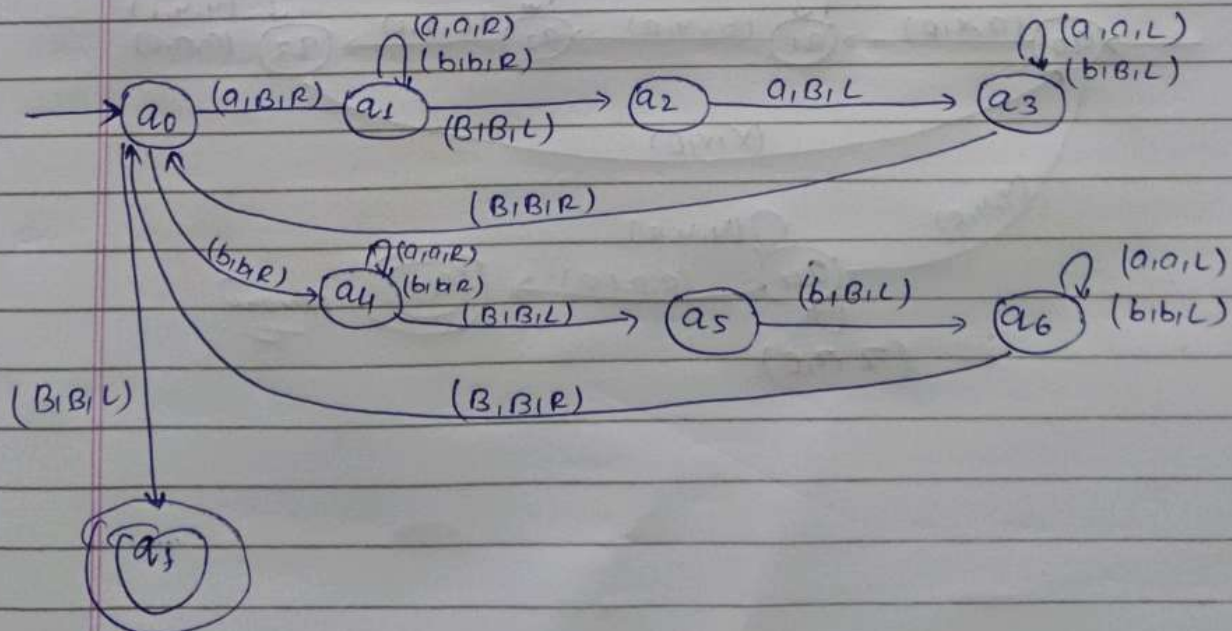
Ans Turing machine have total 7 tuple which is following:-

- Q :- contains a finite set of state.
- T :- A set of input tape.
- B :- It is a blank symbol.
- Σ :- A set of input alphabets.
- δ :- it is a transition function which is $Q \times T \rightarrow Q \times T \times \{L, R\}$
- q_0 :- it is the initial state
- F :- A set of final state.

$$\{ L = wwr \mid w = (a+b)^* \}$$

$$w = ababbaba$$

$\begin{array}{cccccccccccc}
\hline
& a & b & a & b & a & b & a & b & a & b & a & b \\
\hline
& a & b & a & b & a & b & a & b & a & b & a & b \\
& \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\
& \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\
& \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\
\hline
& b & b & b & b & b & b & b & b & b & b & b & b \\
\hline
\end{array}$



Q3 Construct a TM that accepts $L = \{a^n b^n c^n \mid n \geq 1\}$.

Ans $L = a^n b^n c^n \quad n \geq 1$

so $w = aabbbc, aabbbccc, \dots$

Input string = aabbbccc

Tape movement:-

B	a	a	b	b	b	c	c	c	B
B	x	a	a	b	b	b	c	c	c
B	x	a	a	y	b	b	c	c	c
B	x	a	a	y	b	b	z	c	c
B	x	x	a	y	b	b	z	c	c
B	x	x	a	y	y	b	z	c	c
B	x	x	a	y	y	b	z	z	c
B	x	x	x	y	y	b	z	z	c
B	x	x	x	y	y	y	z	z	c
B	x	x	x	y	y	y	z	z	z

